Table 2. Critical values of the t-distribution

α / 2	0,5	0,495	0,475	0,45	0,25	0,2	0,15	0,1	0,05	0,025	0,01	0,005	0,001	0,0005
α	**1**	**0,99**	**0,95**	**0,9**	**0,5**	**0,4**	**0,3**	**0,2**	**0,1**	**0,05**	**0,02**	**0,01**	**0,002**	**0,001**
df(n-1)														
1	0,000	0,016	0,079	0,158	1,000	1,376	1,963	3,078	6,314	12,706	31,821	63,657	318,309	636,619
2	0,000	0,014	0,071	0,142	0,816	1,061	1,386	1,886	2,920	4,303	6,965	9,925	22,327	31,599
3	0,000	0,014	0,068	0,137	0,765	0,978	1,250	1,638	2,353	3,182	4,541	5,841	10,215	12,924
4	0,000	0,013	0,067	0,134	0,741	0,941	1,190	1,533	2,132	2,776	3,747	4,604	7,173	8,610
5	0,000	0,013	0,066	0,132	0,727	0,920	1,156	1,476	2,015	2,571	3,365	4,032	5,893	6,869
6	0,000	0,013	0,065	0,131	0,718	0,906	1,134	1,440	1,943	2,447	3,143	3,707	5,208	5,959
7	0,000	0,013	0,065	0,130	0,711	0,896	1,119	1,415	1,895	2,365	2,998	3,499	4,785	5,408
8	0,000	0,013	0,065	0,130	0,706	0,889	1,108	1,397	1,860	2,306	2,896	3,355	4,501	5,041
9	0,000	0,013	0,064	0,129	0,703	0,883	1,100	1,383	1,833	2,262	2,821	3,250	4,297	4,781
10	0,000	0,013	0,064	0,129	0,700	0,879	1,093	1,372	1,812	2,228	2,764	3,169	4,144	4,587
11	0,000	0,013	0,064	0,129	0,697	0,876	1,088	1,363	1,796	2,201	2,718	3,106	4,025	4,437
12	0,000	0,013	0,064	0,128	0,695	0,873	1,083	1,356	1,782	2,179	2,681	3,055	3,930	4,318
13	0,000	0,013	0,064	0,128	0,694	0,870	1,079	1,350	1,771	2,160	2,650	3,012	3,852	4,221
14	0,000	0,013	0,064	0,128	0,692	0,868	1,076	1,345	1,761	2,145	2,624	2,977	3,787	4,140
15	0,000	0,013	0,064	0,128	0,691	0,866	1,074	1,341	1,753	2,131	2,602	2,947	3,733	4,073
16	0,000	0,013	0,064	0,128	0,690	0,865	1,071	1,337	1,746	2,120	2,583	2,921	3,686	4,015
17	0,000	0,013	0,064	0,128	0,689	0,863	1,069	1,333	1,740	2,110	2,567	2,898	3,646	3,965
18	0,000	0,013	0,064	0,127	0,688	0,862	1,067	1,330	1,734	2,101	2,552	2,878	3,610	3,922
19	0,000	0,013	0,064	0,127	0,688	0,861	1,066	1,328	1,729	2,093	2,539	2,861	3,579	3,883
20	0,000	0,013	0,063	0,127	0,687	0,860	1,064	1,325	1,725	2,086	2,528	2,845	3,552	3,850
21	0,000	0,013	0,063	0,127	0,686	0,859	1,063	1,323	1,721	2,080	2,518	2,831	3,527	3,819
22	0,000	0,013	0,063	0,127	0,686	0,858	1,061	1,321	1,717	2,074	2,508	2,819	3,505	3,792
23	0,000	0,013	0,063	0,127	0,685	0,858	1,060	1,319	1,714	2,069	2,500	2,807	3,485	3,768
24	0,000	0,013	0,063	0,127	0,685	0,857	1,059	1,318	1,711	2,064	2,492	2,797	3,467	3,745
25	0,000	0,013	0,063	0,127	0,684	0,856	1,058	1,316	1,708	2,060	2,485	2,787	3,450	3,725
26	0,000	0,013	0,063	0,127	0,684	0,856	1,058	1,315	1,706	2,056	2,479	2,779	3,435	3,707
27	0,000	0,013	0,063	0,127	0,684	0,855	1,057	1,314	1,703	2,052	2,473	2,771	3,421	3,690
28	0,000	0,013	0,063	0,127	0,683	0,855	1,056	1,313	1,701	2,048	2,467	2,763	3,408	3,674
29	0,000	0,013	0,063	0,127	0,683	0,854	1,055	1,311	1,699	2,045	2,462	2,756	3,396	3,659
30	0,000	0,013	0,063	0,127	0,683	0,854	1,055	1,310	1,697	2,042	2,457	2,750	3,385	3,646
31	0,000	0,013	0,063	0,127	0,682	0,853	1,054	1,309	1,696	2,040	2,453	2,744	3,375	3,633
32	0,000	0,013	0,063	0,127	0,682	0,853	1,054	1,309	1,694	2,037	2,449	2,738	3,365	3,622
33	0,000	0,013	0,063	0,127	0,682	0,853	1,053	1,308	1,692	2,035	2,445	2,733	3,356	3,611
34	0,000	0,013	0,063	0,127	0,682	0,852	1,052	1,307	1,691	2,032	2,441	2,728	3,348	3,601
35	0,000	0,013	0,063	0,127	0,682	0,852	1,052	1,306	1,690	2,030	2,438	2,724	3,340	3,591
36	0,000	0,013	0,063	0,127	0,681	0,852	1,052	1,306	1,688	2,028	2,434	2,719	3,333	3,582
37	0,000	0,013	0,063	0,127	0,681	0,851	1,051	1,305	1,687	2,026	2,431	2,715	3,326	3,574
38	0,000	0,013	0,063	0,127	0,681	0,851	1,051	1,304	1,686	2,024	2,429	2,712	3,319	3,566
39	0,000	0,013	0,063	0,126	0,681	0,851	1,050	1,304	1,685	2,023	2,426	2,708	3,313	3,558
40	0,000	0,013	0,063	0,126	0,681	0,851	1,050	1,303	1,684	2,021	2,423	2,704	3,307	3,551
41	0,000	0,013	0,063	0,126	0,681	0,850	1,050	1,303	1,683	2,020	2,421	2,701	3,301	3,544
42	0,000	0,013	0,063	0,126	0,680	0,850	1,049	1,302	1,682	2,018	2,418	2,698	3,296	3,538
43	0,000	0,013	0,063	0,126	0,680	0,850	1,049	1,302	1,681	2,017	2,416	2,695	3,291	3,532
44	0,000	0,013	0,063	0,126	0,680	0,850	1,049	1,301	1,680	2,015	2,414	2,692	3,286	3,526
45	0,000	0,013	0,063	0,126	0,680	0,850	1,049	1,301	1,679	2,014	2,412	2,690	3,281	3,520
46	0,000	0,013	0,063	0,126	0,680	0,850	1,048	1,300	1,679	2,013	2,410	2,687	3,277	3,515
47	0,000	0,013	0,063	0,126	0,680	0,849	1,048	1,300	1,678	2,012	2,408	2,685	3,273	3,510
48	0,000	0,013	0,063	0,126	0,680	0,849	1,048	1,299	1,677	2,011	2,407	2,682	3,269	3,505
49	0,000	0,013	0,063	0,126	0,680	0,849	1,048	1,299	1,677	2,010	2,405	2,680	3,265	3,500
50	0,000	0,013	0,063	0,126	0,679	0,849	1,047	1,299	1,676	2,009	2,403	2,678	3,261	3,496
51	0,000	0,013	0,063	0,126	0,679	0,849	1,047	1,298	1,675	2,008	2,402	2,676	3,258	3,492
52	0,000	0,013	0,063	0,126	0,679	0,849	1,047	1,298	1,675	2,007	2,400	2,674	3,255	3,488
53	0,000	0,013	0,063	0,126	0,679	0,848	1,047	1,298	1,674	2,006	2,399	2,672	3,251	3,484
54	0,000	0,013	0,063	0,126	0,679	0,848	1,046	1,297	1,674	2,005	2,397	2,670	3,248	3,480
55	0,000	0,013	0,063	0,126	0,679	0,848	1,046	1,297	1,673	2,004	2,396	2,668	3,245	3,476
56	0,000	0,013	0,063	0,126	0,679	0,848	1,046	1,297	1,673	2,003	2,395	2,667	3,242	3,473
57	0,000	0,013	0,063	0,126	0,679	0,848	1,046	1,297	1,672	2,002	2,394	2,665	3,239	3,470
58	0,000	0,013	0,063	0,126	0,679	0,848	1,046	1,296	1,672	2,002	2,392	2,663	3,237	3,466
59	0,000	0,013	0,063	0,126	0,679	0,848	1,046	1,296	1,671	2,001	2,391	2,662	3,234	3,463
60	0,000	0,013	0,063	0,126	0,679	0,848	1,045	1,296	1,671	2,000	2,390	2,660	3,232	3,460
70	0,000	0,013	0,063	0,126	0,678	0,847	1,044	1,294	1,667	1,994	2,381	2,648	3,211	3,435
80	0,000	0,013	0,063	0,126	0,678	0,846	1,043	1,292	1,664	1,990	2,374	2,639	3,195	3,416
90	0,000	0,013	0,063	0,126	0,677	0,846	1,042	1,291	1,662	1,987	2,368	2,632	3,183	3,402
100	0,000	0,013	0,063	0,126	0,677	0,845	1,042	1,290	1,660	1,984	2,364	2,626	3,174	3,390
∞	0,000	0,013	0,063	0,126	0,674	0,842	1,036	1,282	1,645	1,960	2,326	2,576	3,090	3,291

Table 3. Critical values of the Chi-square

α	0,995	0,99	0,975	0,95	0,9	0,5	0,2	0,1	0,05	0,025	0,01	0,005	0,001
df(n-1)													
1	0,000	0,000	0,001	0,004	0,016	0,455	1,642	2,706	3,841	5,024	6,635	7,879	10,828
2	0,010	0,020	0,051	0,103	0,211	1,386	3,219	4,605	5,991	7,378	9,210	10,597	13,816
3	0,072	0,115	0,216	0,352	0,584	2,366	4,642	6,251	7,815	9,348	11,345	12,838	16,266
4	0,207	0,297	0,484	0,711	1,064	3,357	5,989	7,779	9,488	11,143	13,277	14,860	18,467
5	0,412	0,554	0,831	1,145	1,610	4,351	7,289	9,236	11,070	12,833	15,086	16,750	20,515
6	0,676	0,872	1,237	1,635	2,204	5,348	8,558	10,645	12,592	14,449	16,812	18,548	22,458
7	0,989	1,239	1,690	2,167	2,833	6,346	9,803	12,017	14,067	16,013	18,475	20,278	24,322
8	1,344	1,646	2,180	2,733	3,490	7,344	11,030	13,362	15,507	17,535	20,090	21,955	26,124
9	1,735	2,088	2,700	3,325	4,168	8,343	12,242	14,684	16,919	19,023	21,666	23,589	27,877
10	2,156	2,558	3,247	3,940	4,865	9,342	13,442	15,987	18,307	20,483	23,209	25,188	29,588
11	2,603	3,053	3,816	4,575	5,578	10,341	14,631	17,275	19,675	21,920	24,725	26,757	31,264
12	3,074	3,571	4,404	5,226	6,304	11,340	15,812	18,549	21,026	23,337	26,217	28,300	32,909
13	3,565	4,107	5,009	5,892	7,042	12,340	16,985	19,812	22,362	24,736	27,688	29,819	34,528
14	4,075	4,660	5,629	6,571	7,790	13,339	18,151	21,064	23,685	26,119	29,141	31,319	36,123
15	4,601	5,229	6,262	7,261	8,547	14,339	19,311	22,307	24,996	27,488	30,578	32,801	37,697
16	5,142	5,812	6,908	7,962	9,312	15,338	20,465	23,542	26,296	28,845	32,000	34,267	39,252
17	5,697	6,408	7,564	8,672	10,085	16,338	21,615	24,769	27,587	30,191	33,409	35,718	40,790
18	6,265	7,015	8,231	9,390	10,865	17,338	22,760	25,989	28,869	31,526	34,805	37,156	42,312
19	6,844	7,633	8,907	10,117	11,651	18,338	23,900	27,204	30,144	32,852	36,191	38,582	43,820
20	7,434	8,260	9,591	10,851	12,443	19,337	25,038	28,412	31,410	34,170	37,566	39,997	45,315
21	8,034	8,897	10,283	11,591	13,240	20,337	26,171	29,615	32,671	35,479	38,932	41,401	46,797
22	8,643	9,542	10,982	12,338	14,041	21,337	27,301	30,813	33,924	36,781	40,289	42,796	48,268
23	9,260	10,196	11,689	13,091	14,848	22,337	28,429	32,007	35,172	38,076	41,638	44,181	49,728
24	9,886	10,856	12,401	13,848	15,659	23,337	29,553	33,196	36,415	39,364	42,980	45,559	51,179
25	10,520	11,524	13,120	14,611	16,473	24,337	30,675	34,382	37,652	40,646	44,314	46,928	52,620
26	11,160	12,198	13,844	15,379	17,292	25,336	31,795	35,563	38,885	41,923	45,642	48,290	54,052
27	11,808	12,879	14,573	16,151	18,114	26,336	32,912	36,741	40,113	43,195	46,963	49,645	55,476
28	12,461	13,565	15,308	16,928	18,939	27,336	34,027	37,916	41,337	44,461	48,278	50,993	56,892
29	13,121	14,256	16,047	17,708	19,768	28,336	35,139	39,087	42,557	45,722	49,588	52,336	58,301
30	13,787	14,953	16,791	18,493	20,599	29,336	36,250	40,256	43,773	46,979	50,892	53,672	59,703
31	14,458	15,655	17,539	19,281	21,434	30,336	37,359	41,422	44,985	48,232	52,191	55,003	61,098
32	15,134	16,362	18,291	20,072	22,271	31,336	38,466	42,585	46,194	49,480	53,486	56,328	62,487
33	15,815	17,074	19,047	20,867	23,110	32,336	39,572	43,745	47,400	50,725	54,776	57,648	63,870
34	16,501	17,789	19,806	21,664	23,952	33,336	40,676	44,903	48,602	51,966	56,061	58,964	65,247
35	17,192	18,509	20,569	22,465	24,797	34,336	41,778	46,059	49,802	53,203	57,342	60,275	66,619
36	17,887	19,233	21,336	23,269	25,643	35,336	42,879	47,212	50,998	54,437	58,619	61,581	67,985
37	18,586	19,960	22,106	24,075	26,492	36,336	43,978	48,363	52,192	55,668	59,893	62,883	69,346
38	19,289	20,691	22,878	24,884	27,343	37,335	45,076	49,513	53,384	56,896	61,162	64,181	70,703
39	19,996	21,426	23,654	25,695	28,196	38,335	46,173	50,660	54,572	58,120	62,428	65,476	72,055
40	20,707	22,164	24,433	26,509	29,051	39,335	47,269	51,805	55,758	59,342	63,691	66,766	73,402
41	21,421	22,906	25,215	27,326	29,907	40,335	48,363	52,949	56,942	60,561	64,950	68,053	74,745
42	22,138	23,650	25,999	28,144	30,765	41,335	49,456	54,090	58,124	61,777	66,206	69,336	76,084
43	22,859	24,398	26,785	28,965	31,625	42,335	50,548	55,230	59,304	62,990	67,459	70,616	77,419
44	23,584	25,148	27,575	29,787	32,487	43,335	51,639	56,369	60,481	64,201	68,710	71,893	78,750
45	24,311	25,901	28,366	30,612	33,350	44,335	52,729	57,505	61,656	65,410	69,957	73,166	80,077
46	25,041	26,657	29,160	31,439	34,215	45,335	53,818	58,641	62,830	66,617	71,201	74,437	81,400
47	25,775	27,416	29,956	32,268	35,081	46,335	54,906	59,774	64,001	67,821	72,443	75,704	82,720
48	26,511	28,177	30,755	33,098	35,949	47,335	55,993	60,907	65,171	69,023	73,683	76,969	84,037
49	27,249	28,941	31,555	33,930	36,818	48,335	57,079	62,038	66,339	70,222	74,919	78,231	85,351
50	27,991	29,707	32,357	34,764	37,689	49,335	58,164	63,167	67,505	71,420	76,154	79,490	86,661
51	28,735	30,475	33,162	35,600	38,560	50,335	59,248	64,295	68,669	72,616	77,386	80,747	87,968
52	29,481	31,246	33,968	36,437	39,433	51,335	60,332	65,422	69,832	73,810	78,616	82,001	89,272
53	30,230	32,018	34,776	37,276	40,308	52,335	61,414	66,548	70,993	75,002	79,843	83,253	90,573
54	30,981	32,793	35,586	38,116	41,183	53,335	62,496	67,673	72,153	76,192	81,069	84,502	91,872
55	31,735	33,570	36,398	38,958	42,060	54,335	63,577	68,796	73,311	77,380	82,292	85,749	93,168
56	32,490	34,350	37,212	39,801	42,937	55,335	64,658	69,919	74,468	78,567	83,513	86,994	94,461
57	33,248	35,131	38,027	40,646	43,816	56,335	65,737	71,040	75,624	79,752	84,733	88,236	95,751
58	34,008	35,913	38,844	41,492	44,696	57,335	66,816	72,160	76,778	80,936	85,950	89,477	97,039
59	34,770	36,698	39,662	42,339	45,577	58,335	67,894	73,279	77,931	82,117	87,166	90,715	98,324
60	35,534	37,485	40,482	43,188	46,459	59,335	68,972	74,397	79,082	83,298	88,379	91,952	99,607
70	43,275	45,442	48,758	51,739	55,329	69,334	79,715	85,527	90,531	95,023	100,425	104,215	112,317
80	51,172	53,540	57,153	60,391	64,278	79,334	90,405	96,578	101,879	106,629	112,329	116,321	124,839
90	59,196	61,754	65,647	69,126	73,291	89,334	101,054	107,565	113,145	118,136	124,116	128,299	137,208
100	67,328	70,065	74,222	77,929	82,358	99,334	111,667	118,498	124,342	129,561	135,807	140,169	149,449